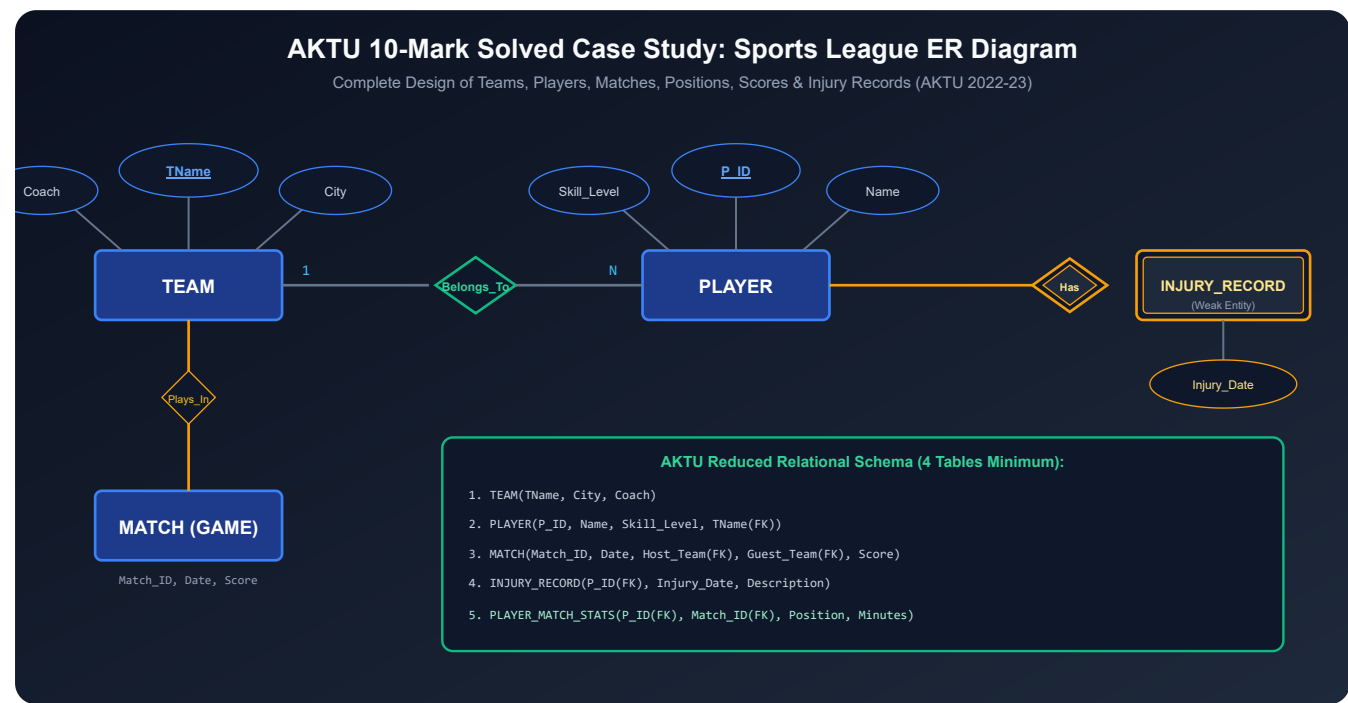


Module 12: Unit 1 AKTU PYQs & Solved Case Studies

Unit 1: Exam Solutions AKTU BCS501 Solved Case Studies

1. Solved 10-Mark Case Study: Sports League ER Design



2. Solved Relational Algebra Queries (AKTU Exam)

Schema: EMPLOYEE(Emp_ID, Name, Salary, Dept_ID), DEPARTMENT(Dept_ID, DName, Location)

1. Find employees in 'Research':
 $\pi_{\text{Name}}(\sigma_{\text{DName} = \text{'Research'}}(\text{EMPLOYEE} \bowtie \text{DEPARTMENT}))$

2. Find employees with Salary > 50,000 in 'Noida':
 $\pi_{\text{Name}}(\sigma_{\text{Salary} > 50000 \wedge \text{Location} = \text{'Noida'}}(\text{EMPLOYEE} \bowtie \text{DEPARTMENT}))$

3. Find departments having ZERO employees:
 $\pi_{\text{DName}}(\text{DEPARTMENT}) - \pi_{\text{DName}}(\text{EMPLOYEE} \bowtie \text{DEPARTMENT})$

3. Core Interview Q&A

Q: Why does Primary Key strictly prohibit NULL?
A: Entity Integrity Constraint demands that every tuple must have an unambiguous, non-null physical identity. In SQL three-valued logic, NULL = NULL evaluates to UNKNOWN, making two NULL keys mathematically indistinguishable.